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INTRODUCTION

Project Context

The motorcycle industry in the Philippines plays an important role in daily transportation, especially in provinces and towns where public transportation is not always available. According to the Federation of Asian Motorcycle Industries as reported by Barro (2025), motorcycle sales in the Philippines reached 1,681,482 units in 2024, eight percent higher than the previous year and the highest recorded in the past five years. As motorcycle ownership grows, so does the demand for parts, accessories, and repair services, placing more pressure on local motorcycle parts shops to manage more products and serve more customers every day. Despite this, small retail shops in the Philippines still rely on paper-based methods such as handwritten logbooks, physical stock counting, and handwritten sales records. A study by Tanaman et al. (2023) found that a small business in Pagadian City, Zamboanga del Sur faced the same problem, where manual recording led to data inaccuracies and inefficient sales reporting, problems that were only resolved after a web-based inventory management system was put in place.

Edmar Motorcycle Parts and Accessories is a local retail shop established in 2009, located in Pandi, Bulacan. The shop sells a wide variety of motorcycle parts and accessories for different motorcycle brands and models, and it also runs a motorcycle repair and maintenance service alongside its retail operations. Both sides of the business operate at the same time and serve different types of customers. Some customers visit only to buy parts and accessories, others come only for repair services, and some need both. The shop is open every day from 6:00 AM to 6:00 PM, except on certain holidays. According to the owners, there are no fixed busy hours since the number of customers changes depending on the day. However, the shop regularly receives between 20 to 30 customers per day, with more customers coming in around the 15th and 30th of each month and during the months of November, December, and January. On the busiest days, the owners shared that up to ten customers may be buying parts while others are having their motorcycles repaired at the same time, which makes it much harder to manage everything without any digital tool to help.

Based on interviews conducted by the proponents with Mr. Edmar and Mrs. Riza Contreras, the owners of the shop, all daily operations are currently handled through handwritten records and manual processes. Inventory is tracked through handwritten logbooks, sales are recorded by hand, and stock levels are checked by physically counting items in the store and storage area from time to time. There is no tool in place that automatically checks stock levels or warns the owners when a product is about to run out. Because of this, stock shortages are only discovered when a customer asks for a specific item and it cannot be found after physically looking for it. The owners also shared that for certain products such as batteries and other items that need to be tested before use, they first check and test the item in front of the customer before the sale is completed, to make sure the product is working before it leaves the shop. For other items that do not require testing, if a customer discovers that a product is damaged after purchase, the shop offers a refund. Damaged items are then returned to the supplier and manually removed from the shop's stock records. Since this process is handled entirely by hand, there is no organized way to track which items were returned, how many were removed from stock, or when the return was made. The proponents used purposive sampling to select 20 customers who had recent buying experience at the shop, so that the feedback gathered would be relevant and based on actual transactions. Of those interviewed, 25% said they had visited the shop and found that the specific part they needed was not available at the time. While the rest of the customers were satisfied with their visit, those who experienced stock unavailability left without getting what they came for. For a shop that depends on walk-in customers from the local community, this kind of problem can lead to lost sales and can discourage customers from coming back.

Another problem the shop regularly faces is answering customer questions about part compatibility. When a customer asks if a specific part fits their motorcycle model, the owners have to rely on memory or go through printed catalogs and folders to find the answer. This takes time and can lead to mistakes, especially when several customers are asking questions at the same time. This challenge is made more difficult by the fact that some motorcycle parts can be used across different brands and models. As explained by ProNorthParts (2024), motorcycle parts can be shared between different brands when the specifications or part numbers match, meaning parts from brands such as Honda,

Yamaha, Suzuki, and Kawasaki can sometimes be used interchangeably depending on the part. Lopez (2024) adds that brake pads, for example, can be used across different motorcycle brands and models as long as the specifications are the same. With a large number of parts and motorcycle models to keep track of, checking compatibility by hand becomes more difficult and less reliable as the shop continues to grow.

On the repair side of the business, the number of customers per day is generally lower, but the time needed to finish a repair job can range from 30 minutes to more than an hour and a half, depending on how complex the job is. With no system to track ongoing repairs, expected finish times, or the parts used for each job, managing the repair side alongside the retail floor becomes harder, especially on the busier days of the month. The owners also shared that they do not have a fixed set of suppliers. They buy their stock from stores and warehouses in Bunsuran, Marilao, and Santol, and also order products through the Shopee app, depending on what is available and how much it costs at the time. Because of this, there is no regular restocking schedule, which makes it even more important for the shop to know exactly how much stock they have before something runs out.

The owners also confirmed that the shop has previously practiced an informal kind of pre-ordering, where they purchase specific parts in advance for regular customers who request them ahead of time, usually with a down payment. Since this is already something the shop has been doing, the owners agreed that having a proper system to record and manage pre-orders with down payment tracking would be a useful addition.

All of these problems taken together show a business that has grown beyond what its current paper-based methods can handle. The problems of undetected stock shortages, inaccurate inventory records, difficulty checking part compatibility, the lack of organized sales and repair job tracking, the absence of a structured pre-order process, and the untracked handling of damaged items and product returns are the key operational challenges that this study aims to address for Edmar Motorcycle Parts and Accessories.

General Problem

The problem of the study is that Edmar Motorcycle Parts and Accessories manages its daily operations through entirely manual and paper-based processes, which makes it difficult for the owners to monitor stock levels accurately, determine part compatibility, and serve customers efficiently.

Specific Problems

- **Real-Time Stock Monitoring**

  The shop has no way of knowing how many items are left in stock unless the owners physically check. Because of this, stock shortages are only discovered at the moment a customer asks for a product.

- **Inventory Management of Motorcycle Parts and Accessories**

  The shop carries a wide variety of parts for different motorcycle makes and models. Without a digital inventory system, all products, quantities, and details are tracked manually through handwritten logbooks and physical counts, which is time-consuming and prone to errors.

- **Part Compatibility Checking**

  The owner currently relies on memory or printed catalogs to answer compatibility questions, which is slow and error-prone especially during busy periods.

- **Point of Sales Transaction Recording**

  Sales are recorded by hand with no centralized record, making it difficult to monitor which products are frequently sold, which items are running low, and how the business is performing overall.

- **Repair and Service Request Management**

  The repair shop sometimes only handles around five customers a day, but with no way to track ongoing and pending jobs, managing the workflow during busier periods still becomes difficult and disorganized.

- **Report Generation**

  All records are paper-based, so the owners have no easy way to review historical data, track stock depletion patterns, or prepare for high-demand periods.

Purpose and Description

The purpose of this project is to develop a web-based motorcycle parts inventory and point of sales system with compatibility checker for Edmar Motorcycle Parts and Accessories to manage its products, sales, and repair services more efficiently. The shop currently handles all of its daily operations through handwritten logbooks, manual stock counting, and paper-based sales recording, which has led to recurring problems such as undetected stock shortages, inaccurate records, and slow customer service. The system will organize all inventory and sales information in one platform, reducing the owners' reliance on handwritten logbooks and physical counts, while providing a faster and more accurate way to serve customers and keep track of daily operations.

A key feature of MOTOTRACK is the compatibility checker, which will allow both the owners and customers to look up whether a specific motorcycle part is compatible with a particular motorcycle make and model. Customers can use this feature even before visiting the shop or making a purchase, so they already know in advance whether the part they need fits their motorcycle. Since certain parts can be used across different brands and models, the compatibility checker is designed to reflect this. The system will also include real-time stock monitoring that will notify the owners when a product is running low, an inventory management module, a point of sales transaction log, a repair and service request tracker, a pre-order feature with down payment for products that are currently out of stock, and a report generation feature for a clear overview of business operations.

Objectives

General Objective

To develop MOTOTRACK, a web-based motorcycle parts inventory and point of sales system with compatibility checker for Edmar Motorcycle Parts and Accessories that automates stock tracking, simplifies part compatibility verification, and organizes sales and repair records through a centralized dashboard.

Specific Objectives

- **To be able to develop a real-time stock monitoring feature**
  This will provide the owners with updates on inventory levels and send automatic alerts.
- **To be able to develop a digital inventory management module**
  This will store and organize all product information, quantities, and details in a centralized database, which makes it faster than the handwritten logbooks and physical counts currently used by the shop.
- **To be able to develop a motorcycle parts compatibility checker**
  This will allow the owners and users to verify if a part fits a specific motorcycle make and model.
- **To be able to develop point of sales feature**
  This will process and record every completed sale and update inventory levels accordingly, replacing hand-recorded entries with accurate and centralized digital records.
- **To be able to develop a service tracking module**
  This will help the shop keep track of ongoing and pending repair jobs, reducing confusion and helping the owners manage the workflow even during busier periods**.**
- **To be able to develop a report generation feature**
  This will give the owners a clear and organized overview of their inventory status and business operations.

Scope and Limitations

Scope

The study covers the development of MOTOTRACK, a web-based motorcycle parts inventory and point of sales system with compatibility checker, specifically designed for Edmar Motorcycle Parts and Accessories in Pandi, Bulacan.

Shop Owner

- Dashboard overview of inventory and sales
- Inventory management of motorcycle parts and accessories
- Parts compatibility checker management
- Point of sales and transaction recording
- Repair and service request tracking
- Pre-order management with down payment recording
- Report generation for stock levels, sales, and business operations

Customer

- Motorcycle parts compatibility checker to verify if a specific part fits a particular motorcycle make and model before visiting the shop
- Online purchase of motorcycle parts and accessories with full payment via GCash
- Pre-order placement for out of stock products with down payment via GCash

Limitations

While MOTOTRACK is designed to address the key operational problems of Edmar Motorcycle Parts and Accessories, the system operates within the following boundaries and constraints:

- The compatibility checker can only verify parts and motorcycle models that have already been recorded in the system by the owners.
- New products, stock quantities, and other details must be entered manually by the owners before the system can begin tracking them.
- The system does not include accounting functions, supplier management, delivery tracking, or payroll processing.
- The system does not have a dedicated mobile application.
- The only accepted payment method for both online purchases and pre-order down payments is GCash.
- For pre-orders, only the down payment amount is recorded. Any additional purchases made at the time of pick-up will be treated as a separate sales transaction.
- The system is designed for single-branch use only and does not support businesses operating across multiple locations.

REVIEW OF RELATED LITERATURE/STUDIES/SYSTEMS

Review of Related Literature

Real-Time Stock Monitoring with Low Stock Alerts

Real-time stock monitoring is the continuous and automated tracking of inventory levels to maintain accurate and up-to-date records across a system. Roy et al. (2025) explain that traditional inventory methods that rely on manual counting and spreadsheet-based tracking are prone to errors, delays, and a lack of real-time visibility, which can lead to stockouts and overstocking. The integration of automated low-stock alerts allows the system to notify users when inventory falls below a set level, removing the need for constant manual checking. This approach minimizes human error and ensures that inventory information is always current, allowing shop owners to respond to stock shortages before they affect operations.

Inventory Management

Inventory management is the systematic process of overseeing and controlling the flow of parts and products to ensure that the right items are available at the right time while keeping costs under control. Puiu et al. (2025) emphasize that effective inventory management is essential in the automotive parts industry due to the growing complexity of supply chains and the constant changes in market demand. Their study used the Six Sigma DMAIC cycle to identify the root causes of inventory inaccuracies and remove activities that do not add value to the process. By focusing on stock accuracy and streamlining how parts are tracked and managed, the research showed that businesses can significantly reduce inventory errors, avoid unnecessary costs, and maintain a steady and reliable supply of parts for their operations.

Parts Compatibility Checker

A parts compatibility checker is a digital tool that allows both shop owners and customers to verify whether a specific part fits a particular vehicle make, model, and year before a purchase is made. Nathan et al. (2024) highlight that one of the most persistent challenges in automotive parts management is ensuring that the correct parts are matched to the

correct vehicle, and that a structured digital framework for managing service parts significantly reduces selection errors, minimizes incorrect orders, and improves overall customer satisfaction. By having a system that filters available parts based on vehicle specifications, shops can prevent customers from purchasing incompatible parts, reduce the risk of returns and complaints, and build greater trust in the reliability of their service.

Point of Sales Transaction Recording

Sales transaction recording is the systematic process of documenting every sale made by a business in a centralized and organized manner to ensure accurate financial and inventory records. Melvin et al. (2023) developed a web-based Point of Sale application for a small automotive spare parts shop in Indonesia and found that the main problem the shop faced was the discrepancy between the actual number of products sold and what appeared in the sales reports due to the lack of an integrated recording system. Their implemented system resolved this by automatically updating inventory records every time a sale was completed, ensuring that stock levels and sales data remained accurate and consistent. The system was tested using the System Usability Scale and received an average score of 86.6, placing it in the very good category, which confirms that integrated sales transaction recording systems are both practical and effective for small automotive parts businesses.

Repair and Service Request Management

Repair and service request management is the process of organizing, tracking, and monitoring vehicle maintenance tasks from the time a customer submits a repair request until the vehicle is returned. Titu and Pop (2025) examined how implementing structured process monitoring systems in automotive repair organizations significantly improves service quality and operational efficiency. Their study, conducted across two automotive service centers over a six-month period, found that applying quality management tools such as Kaizen and Lean Manufacturing reduced rework rates from 7.8% to 2.6%, improved technician efficiency from 89% to 105%, and shortened average service completion time by 1.6 days. These findings demonstrate that when repair and service

processes are properly tracked and managed through a digital system, workshops can deliver faster, more reliable, and more transparent service to their customers.

Report Generation

Report generation is the automated process of compiling business data into organized summaries that help owners and managers monitor performance and make informed decisions. Artene et al. (2024) examined how integrating digital reporting tools into business systems provides organizations with real-time financial insights and data-driven decision-making capabilities. Their study found that automated reporting systems remove the need for manual data compilation, reduce the risk of errors, and allow business owners to quickly identify trends and make adjustments based on current performance data. For small businesses in particular, having access to clear and regularly generated reports on stock levels and product movement is essential for maintaining operational control and planning future purchases effectively.

Pre-Order

A pre-order feature allows customers to reserve and pay a deposit for products that are not currently available in stock, giving the business a committed order before the item arrives. Puengwattanapong and Leelasantitham (2022) studied the behavior of online consumers across three stages of the purchasing process, namely pre-purchase, purchase, and post-purchase, and found that the pre-purchase stage is a critical point where consumers make decisions about whether to commit to a product before it becomes fully available. Their study found that when consumers are given the opportunity to engage with a product early through a structured pre-order process, their intention to follow through with the purchase increases, providing the business with guaranteed demand and reducing the guesswork involved in procurement. For a motorcycle parts shop, a structured pre-order feature with down payment recording makes the process more reliable and easier to manage for both the owners and the customers.

Benchmarking

Features	Torque360	MCA Bike	Setmore	RevZilla	Webike Philippines	MOTOTRACK: The Development of a Web-Based Motorcycle Parts Sales and Inventory Management System with Compatibility Checker for Edmar Motorcycle Parts and Accessories, Pandi, Bulacan
1. Real-Time Stock Monitoring with Low Stock Alerts		✓		✓		✓
2. Inventory Management of Motorcycle Parts and Accessories	✓	✓		✓	✓	✓
3. Parts Compatibility Checker				✓	✓	✓
4. Pont of Sales Transaction Recording	✓	✓	✓			✓
5. Repair and Service Request Management	✓	✓	✓			✓
6. Report Generation	✓	✓	✓			✓
7. Pre-Order				✓	✓	✓

Table 1.0 Benchmarking Table

Among the five existing systems reviewed, inventory management of motorcycle parts and accessories is the most widely supported feature, present in four of them: Torque360, MCA Bike, RevZilla, and Webike Philippines. This shows that organizing and displaying products is a standard function across both local and foreign platforms. Real-time stock monitoring with low stock alerts, however, is only available in two of the five systems, which are MCA Bike and RevZilla. Point of Sales transaction recording is supported by Torque360, MCA Bike, and Setmore, but is not found in RevZilla and Webike Philippines. Repair and service request management is also only offered by Torque360, MCA Bike, and Setmore. Report generation is found in four of the five systems, specifically Torque360, MCA Bike, Setmore, and Webike Philippines. Pre-order functionality is available in RevZilla and Webike Philippines.

The parts compatibility checker is the least common feature among the systems reviewed, with only RevZilla and Webike Philippines offering it. However, neither of these two platforms includes point of sales transaction recording or repair and service request management. This means that none of the existing systems reviewed combines a parts compatibility checker, point of sales recording, and service management all in one platform. These features represent the most important needs identified through direct interviews with the owners and customers of Edmar Motorcycle Parts and Accessories.

MOTOTRACK addresses all of these gaps by bringing all seven features together in one platform built specifically for a small local motorcycle parts and repair shop. Torque360, MCA Bike, and Setmore each cover point of sales recording, repair management, and report generation, but none of them includes a parts compatibility checker or pre-order functionality. RevZilla offers the compatibility checker and pre-order options but does not have point of sales transaction recording, repair tracking, and stock alerts. Webike Philippines provides the compatibility checker, pre-order support, and report generation, but does not include point of sales recording, repair tracking, and stock monitoring. By putting all seven features into a single system designed around the actual needs of Edmar Motorcycle Parts and Accessories, MOTOTRACK offers a more complete and practical solution compared to all the existing systems reviewed.

Related Studies and/or Systems

Torque360

Torque360 is a specialized repair shop management platform engineered for independent motorcycle mechanics. It excels in Repair and Service Request Management through interactive digital job cards and a dedicated technician portal for tracking labor hours. It also handles Sales Transaction Recording and Report Generation, allowing shop owners to process payments and view detailed earnings summaries.

MCA Bike

MCA Bike is a modular, cloud-based ERP solution specifically developed for motorcycle and bicycle workshops. It features a robust Real-Time Stock Monitoring system with automated low-stock alerts and advanced Inventory Management that supports multi-location tracking. The system streamlines business operations by integrating Sales Transaction Recording, Repair and Service Request Management, and Report Generation into a single interface.

Setmore

Setmore is a cloud-based scheduling platform used by independent motorcycle mechanics to automate their booking workflows. It is designed to be a lean service tool that focuses on Repair and Service Request Management through a 24/7 self-booking page, allowing customers to book appointments at any time without needing to call the shop directly. Shop owners can view and manage all bookings through a centralized calendar, making it easier to organize daily repair schedules. It also handles Sales Transaction Recording by collecting payments through the platform and provides Report Generation by tracking total appointments and revenue, giving shop owners a basic overview of their business performance.

RevZilla

RevZilla is a leading global e-commerce platform for motorcycle parts and accessories, renowned for its highly accurate "Browse-by-Bike" digital fitment tool. This feature

functions as a comprehensive Parts Compatibility Checker, allowing users to filter thousands of products by specific year, make, and model to ensure exact fitment before purchase. While it excels in retail inventory and pre-order capabilities for aftermarket gear, RevZilla is purely a consumer-facing store. It does not provide administrative tools for shop owners, such as Repair and Service Request Management, technician scheduling, or internal Sales Transaction Recording for a physical storefront.

Webike Philippines

Webike Philippines is the Philippine branch of Webike Japan, recognized as Japan's number one online store for motorcycle parts and accessories. It operates through its website at webike.ph with offices in Parañaque and Mandaue City, Cebu, and carries over two million products from major motorcycle brands including Honda, Suzuki, Kawasaki, Yamaha, Ducati, BMW, and KTM. One of its key features is the Mybike service, which allows registered users to input their motorcycle make, model, and year so the platform can display only compatible parts. It also offers a quotation and pre-order feature for parts that are not currently in stock, where customers submit a request and Webike sources and notifies them once the product is available.

Synthesis

The five related systems reviewed in this study show that digital platforms for motorcycle parts and repair services are already being used in the industry. Torque360 and MCA Bike demonstrate that combining inventory management, repair tracking, point of sales transaction recording, and report generation in one system is both possible and practical. Setmore shows that even a simple scheduling tool can help small shops manage service requests more efficiently. RevZilla and Webike Philippines show that customers are already comfortable using digital tools to browse parts, check compatibility, and place pre-orders when items are out of stock.

Despite this, none of the five systems are able to meet all the needs of a small locally-owned motorcycle parts and repair shop in the Philippines. Torque360, MCA Bike, and Setmore are strong in repair and point of sales management but do not include a parts compatibility checker or pre-order functionality. RevZilla and Webike Philippines offer these two features but are built purely as online stores and lack the back-end tools that a physical shop needs, such as point of sales transaction recording, repair tracking, and report generation. Most importantly, none of the five systems bring together a parts compatibility checker, point of sales transaction recording, and repair management in one platform, which are among the most critical needs identified through direct interviews with the owners and customers of Edmar Motorcycle Parts and Accessories.

These gaps show that while each existing system addresses certain parts of the problem, none of them offer a complete solution for a small local motorcycle parts and repair shop. MOTOTRACK aims to fill this gap by combining all seven features in one platform: real-time stock monitoring, inventory management, a parts compatibility checker, point of sales transaction recording, repair and service request tracking, a pre-order feature, and report generation. These features are not added for the sake of adding them, but are directly based on the real and documented needs of Edmar Motorcycle Parts and Accessories, making MOTOTRACK a necessary and practical proposed system.

TECHNICAL BACKGROUND

Conceptual Model of the Study

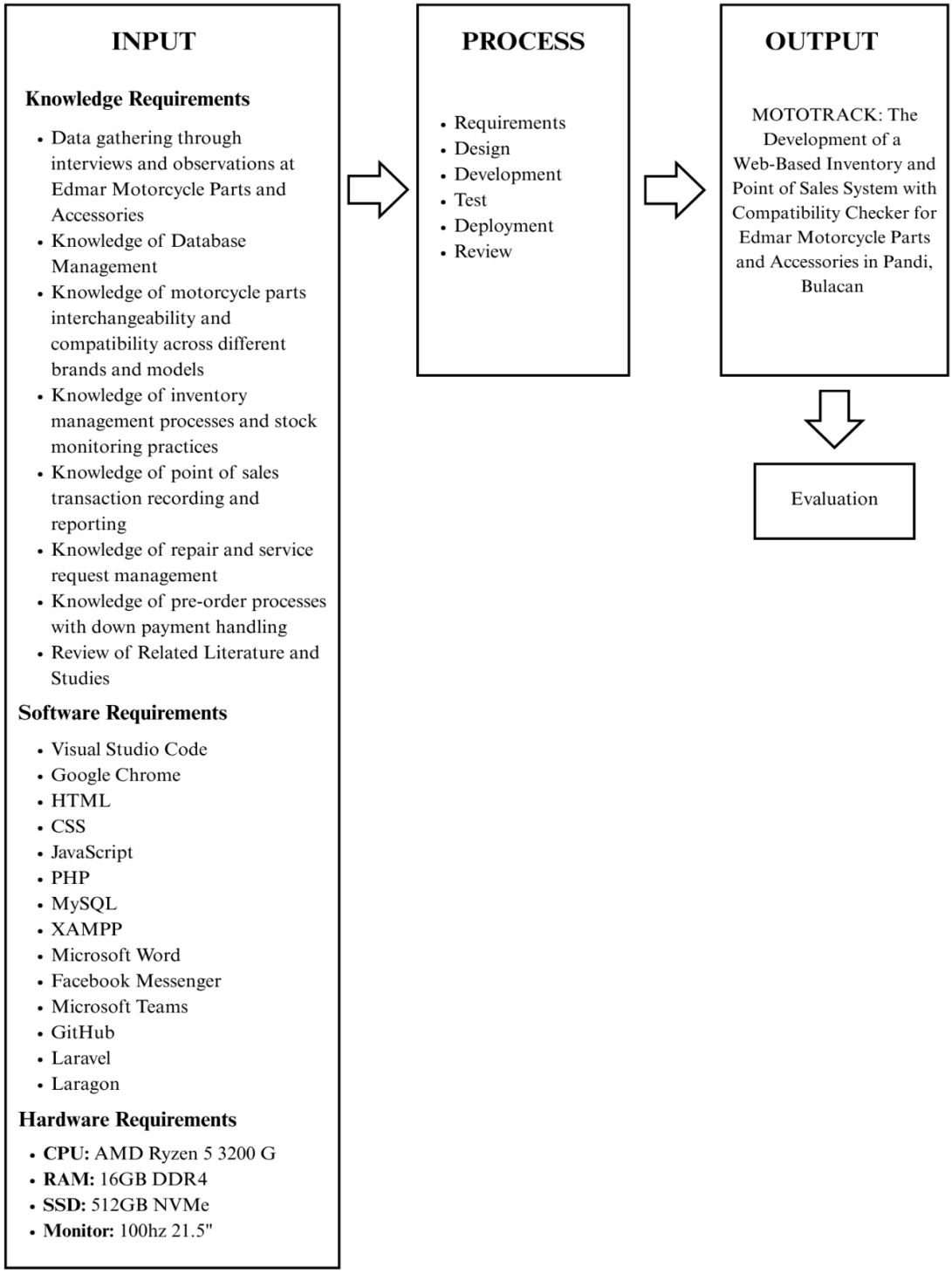


Figure 1.0 Input-Process-Output (IPO) Framework

The IPO framework illustrates the systematic approach used in developing MOTOTRACK, a web-based inventory and point of sales system with compatibility checker for Edmar Motorcycle Parts and Accessories in Pandi, Bulacan.

The INPUT section is divided into three parts. The first is the knowledge requirements, which covers the information gathered through interviews and observations at the shop, including the key operational problems such as undetected stock shortages, manual inventory tracking, part compatibility checking, point of sales transaction recording, repair and service request management, and pre-ordering with down payment. Also included are the knowledge of database management, motorcycle parts interchangeability, and a review of related literature and studies. The second is the software requirements, which lists the development tools and web technologies to be used in building the system. The third is the hardware requirements, which specifies the equipment needed for the system to run properly. Together, these inputs make sure that the needs of the shop owners and customers are properly documented and that all technical requirements are identified before development begins.

The PROCESS follows the standard phases of software development, starting from requirements gathering, followed by design, development, testing, deployment, and review. Each phase is completed before moving to the next, while still allowing for adjustments based on feedback gathered during testing. This structured approach ensures that every feature of the system is built and tested properly before the final system is delivered.

The OUTPUT is MOTOTRACK, a fully functional web-based inventory and point of sales system with compatibility checker designed specifically for Edmar Motorcycle Parts and Accessories. Once delivered, the system is then subjected to evaluation to determine whether it meets all the objectives of the study and provides measurable improvements over the current manual processes of the shop.

- **Technologies to be Used**

The development of MOTOTRACK involves several tools and technologies that work together to build a complete and functional web-based system. Each tool was chosen based on the specific needs of the project and the capabilities required to address the manual process problems of Edmar Motorcycle Parts and Accessories.

Visual Studio Code shall be used as the main code editor throughout the development process. It is free, beginner-friendly, and supports the languages and frameworks used in the project, making it a practical choice for the entire development process.

The front end of MOTOTRACK shall be built using HTML, CSS, and JavaScript, where HTML defines the structure of each page, CSS handles the visual design, and JavaScript manages the interactive features of the system. On the server side, PHP shall handle the back-end logic, processing user requests and communicating with the database to deliver the appropriate responses. Laravel shall serve as the PHP framework, following the Model-View-Controller pattern to organize the system's logic, database interactions, and interface into manageable components, while also providing built-in tools for routing, authentication, and form validation.

MySQL shall be used to store and manage all data in the system, including product information, inventory levels, sales transactions, repair requests, compatibility data, and pre-order details. XAMPP and Laragon shall provide the local server environments needed to run and test the system throughout development. XAMPP offers a straightforward setup that includes Apache, MySQL, and PHP, while Laragon serves as a modern and portable alternative that supports the same core technologies with faster performance, a cleaner interface, and additional tools such as Composer and Git, making both environments well-suited for developing and testing a Laravel-based application. GitHub shall be used for version control, allowing the team to track changes and maintain a secure backup of the source code. Google Chrome shall serve as the primary browser for testing the system's interface and functionality during development.

Calendar of Activities

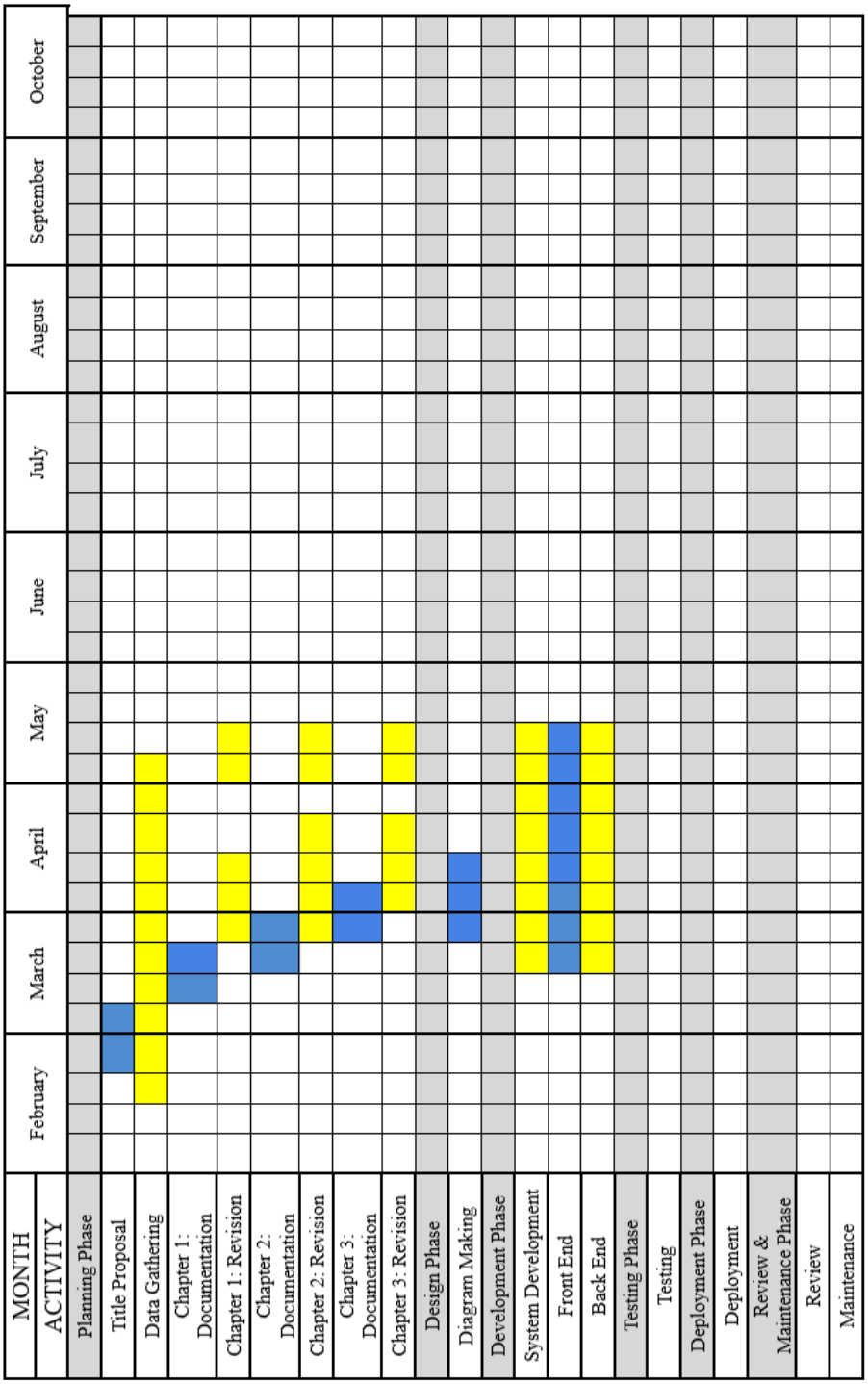


Table 2.0 Gantt Chart of Activities

- **Resources**

  Hardware

Hardware Components	Hardware Specification
Operating System	Windows 11
Solid State Drive (SSD)	512 GB NVMe
Random Access Memory (RAM)	16GB DDR4
Processor & Graphics Processing Unit	AMD Ryzen 5 3200 G
Peripherals	Monitor, Keyboard, and Mouse

Table 3. Hardware Resources

  Software

Software Categories	Specific Software
Integrated Development Environment	Visual Studio Code
Web Development Technologies	HTML, CSS, JavaScript, PHP
Database	MySQL
Local Server Environment	XAMPP, Laragon
Communication Tool	Facebook Messenger, Microsoft Teams
Word Processing Software	Microsoft Word
Internet Browser	Google Chrome
Framework	Laravel
Hosting Platform	GitHub

Table 4. Software Resources

Visual Studio Code

Visual Studio Code, commonly known as VS Code, is the main code editor used by the proponents throughout the development of MOTOTRACK. It is a free and lightweight software developed by Microsoft that supports the web development languages used in the project. Its built-in features such as error highlighting, code suggestions, and an integrated terminal help make the development process more organized and efficient. VS Code was chosen because it is practical, easy to work with, and well-suited for the type of web-based system the proponents are building.

Web Development Technologies

The front end of MOTOTRACK is built using three core web technologies: HTML, CSS, and JavaScript. HTML is used to define the structure and layout of each page in the system. CSS controls the visual appearance of the interface, including the design, spacing, and overall presentation of each page. JavaScript handles the interactive and dynamic parts of the system, such as the motorcycle parts compatibility checker, inventory dashboard, and sales transaction forms. These three technologies work together to produce an interface that is easy to navigate and accessible through any standard web browser. On the server side, PHP is used to handle the back-end logic of the system. It processes user requests, communicates with the MySQL database, and generates the appropriate responses for each page. PHP works alongside Laravel as its underlying language, allowing the proponents to build a fully functional and data-driven web application that connects the front-end interface with the database efficiently.

MySQL

MySQL is the database management system used to store and manage all data in MOTOTRACK. It is a free and widely used relational database that organizes information into structured tables, making it efficient for storing and retrieving large amounts of data. MySQL handles all records in the system, including product information, inventory levels, sales transactions, repair and service requests, compatibility data, and pre-order details. It was selected because of its reliability, its

compatibility with web-based applications, and its ability to keep data accurate and consistent throughout daily operations.

Local Server Environment

XAMPP and Laragon are the local server environments used by the proponents to develop and test MOTOTRACK before deployment. Both are free and open-source software packages that provide essential components for web development. XAMPP includes Apache as the web server, MySQL for the database, and PHP for server-side scripting, while Laragon offers a modern, portable alternative that also supports Apache or Nginx, MySQL, and PHP, along with additional tools such as Composer, Git, and Node.js for a more streamlined development workflow. These environments allowed the proponents to run and test the web application directly on their local machines without requiring an internet connection or a live server. They were chosen because both are easy to set up, widely used in academic and web development settings, and fully compatible with the technologies used in building MOTOTRACK. Laragon, in particular, is favored for its fast performance, clean interface, and ability to quickly switch between different versions of PHP and other components.

Communication Tool

Facebook Messenger and Microsoft Teams are the communication platforms used by the proponents throughout the development of MOTOTRACK. Facebook Messenger was primarily used for day-to-day communication among the members of the team, including sending updates, coordinating tasks, and sharing files. Microsoft Teams was used specifically for communication with their research adviser, including online meetings, document sharing, and receiving feedback on the progress of the study. Both platforms were chosen because they are free, readily available, and already familiar to all members of the group.

Microsoft Word

Microsoft Word is the word processing software used by the proponents to write, edit, and format all documents related to the study. This includes the written chapters of the capstone project, tables, and other formal documents required throughout the research process. It was used because of its wide availability and its formatting tools that are appropriate for producing academic and technical documents.

Google Chrome

Google Chrome is the internet browser used by the proponents as the primary testing environment for MOTOTRACK during development. It was used to view the system's interface and check how features look and function in a real browser setting as changes were made. Its built-in developer tools, including the element inspector and console, also helped the proponents identify and correct layout issues and errors throughout the development process.

Laravel

Laravel is the PHP framework used by the proponents for building the back-end structure of MOTOTRACK. It follows the Model-View-Controller (MVC) architectural pattern, which helps organize the system's logic, database interactions, and user interface into separate, manageable components. Laravel provides built-in tools for routing, authentication, database migration, and form validation, which significantly reduced development time and minimized errors. It was chosen because of its clean and readable syntax, its well-structured approach to web application development, and its strong compatibility with MySQL and other tools used in the project.

GitHub

GitHub is the cloud-based platform used by the proponents for version control and source code management. Integrated with Git and VS Code, it allows the team to track changes, collaborate remotely, and maintain an online backup of the source code. It will also be used for future maintenance and updates following the completion of the project.

METHODOLOGY

The proponents shall use the Agile Development Methodology for developing MOTOTRACK: a Web-Based Inventory and Point of Sales System with Compatibility Checker for Edmar Motorcycle Parts and Accessories, Pandi, Bulacan.

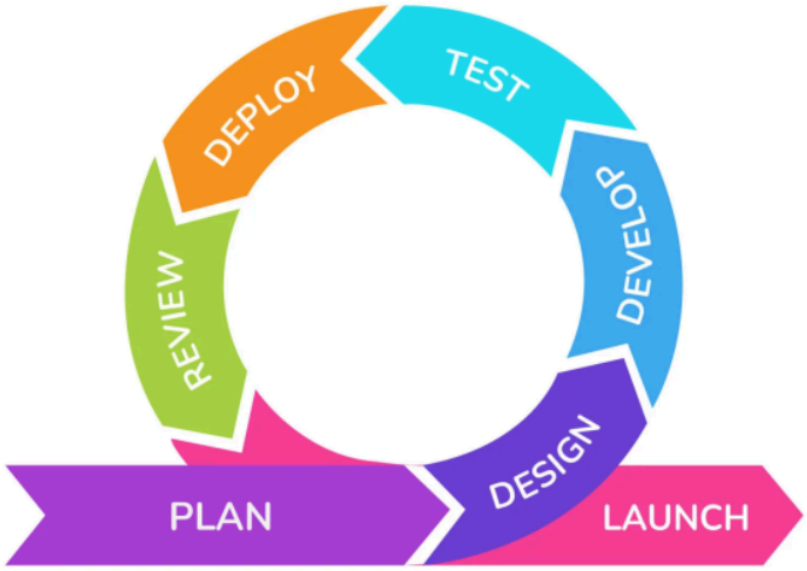


Figure 2.0 Agile Methodology

Planning

In the first week of March, the proponents conducted an initial interview with the owners of Edmar Motorcycle Parts and Accessories to gather the necessary information about the shop's current manual processes and the problems they experience in their daily operations. Through this initial data gathering activity, the proponents identified the specific needs of the shop and used these findings as the basis for proposing MOTOTRACK as a solution. In the second week of March, the proponents returned to the shop to conduct a follow-up interview with the owners to gather additional details and clarify information that was needed to further refine the scope and features of the system. All collected information was documented and reviewed to ensure that the system being developed directly addresses the problems of Edmar Motorcycle Parts and Accessories.

Design

Once the necessary data had been gathered, the proponents began planning the overall structure and layout of MOTOTRACK. Low-fidelity prototype sketches of the system's interface were created first to map out the general layout and flow before moving on to more detailed design work. After the sketches were finalized, the proponents proceeded to build the front end of the system using HTML, CSS, and JavaScript in Visual Studio Code, where layouts and components such as buttons, input fields, navigation bars, and tables were created and styled. XAMPP and Laragon were set up during this phase to provide local server environments, allowing the proponents to run and test the system directly on their machines. Google Chrome was used to view and review the system's interface in real time as changes were made. Flowcharts and data flow diagrams were also developed alongside the design process to clearly visualize how the system's features and processes would work, including inventory management, the parts compatibility checker, sales transaction recording, repair tracking, pre-order management, and report generation. These design outputs served as the guide for the development phase to ensure that the final system is organized, logical, and easy to navigate.

Development

With the designs finalized, the proponents will proceed to building MOTOTRACK using HTML, CSS, and JavaScript for the front end, PHP for the server-side logic, and MySQL for the database. Laravel will be used as the primary PHP framework to organize the back-end structure of the system following the Model-View-Controller pattern, which separates the system's logic, database interactions, and interface into manageable components.XAMPP and Laragon will continue to serve as the local server environments throughout this phase, allowing the proponents to run and test the system on their local machines. Visual Studio Code will be used as the primary development environment throughout this phase. All core features of the system will be developed, including real-time stock monitoring, inventory management, the parts compatibility checker, sales transaction recording, repair and service request tracking, pre-order management with down payment recording, and report generation.

Testing

After development, the proponents will thoroughly test every feature of MOTOTRACK to ensure that all parts of the system are working correctly and as intended. Both the front end and back end will be checked to confirm that they are properly connected and functioning together, including the PHP and Laravel back-end processes and their communication with the MySQL database through XAMPP and Laragon. The shop owners will also be involved in the testing process by trying out the system firsthand and providing feedback. Any errors or issues identified during this phase will be corrected before the system proceeds to deployment.

Deployment

Once the system has passed all tests and has been confirmed to be fully functional, MOTOTRACK will be deployed and made accessible through a web browser for use at Edmar Motorcycle Parts and Accessories. The proponents will ensure that the system is properly set up and that all data and configurations are in place before the owners begin using it for daily operations. The goal of this phase is to ensure a smooth and problem-free transition from the current manual processes to the new digital system.

Review

After deployment, the proponents will monitor the performance of MOTOTRACK and gather feedback from the shop owners based on actual use. Any issues or areas that need improvement will be identified and addressed through the necessary updates and adjustments. This phase will also allow the proponents to evaluate whether the system has successfully met all of its objectives and whether any additional improvements can be made to further enhance the system's usefulness for the shop.

Requirements Analysis

Who: The main users of the MOTOTRACK system will include the following:

Shop Owner – The primary and only authorized user of the administrative side of the system. The owners are responsible for managing all daily operations of Edmar Motorcycle Parts and Accessories, including monitoring stock levels, managing inventory, recording point of sales transactions, tracking repair and service requests, processing pre-orders with down payment, and generating business reports.

Customers – Individuals who visit or inquire at the shop to purchase motorcycle parts and accessories or avail of repair services. They can use the compatibility checker to verify whether a specific part fits their motorcycle make and model before making a purchase, purchase motorcycle parts and accessories online with full payment via GCash, and place pre-orders for out of stock products with down payment via GCash.

What - The business focuses on selling motorcycle parts and accessories and providing repair and maintenance services for various motorcycle makes and models. The shop currently relies on manual and paper-based processes such as handwritten logbooks and physical inventory counts, which often lead to undetected stock shortages, compatibility errors, and disorganized records. MOTOTRACK aims to automate and improve these processes by providing the following features: real-time stock monitoring with low-stock alerts, digital inventory management, a motorcycle parts compatibility checker, point of sales transaction recording, repair and service request tracking, a pre-order feature with down payment recording, and report generation for a clear overview of business operations.

Where - MOTOTRACK will primarily be used at Edmar Motorcycle Parts and Accessories in Pandi, Bulacan, where the shop owners will manage all operations through the system's central dashboard. Since MOTOTRACK is a web-based application, it can also be accessed through any internet browser on a mobile phone, laptop, or desktop computer. Customers can use the compatibility checker and online purchasing features

from any location with an internet connection, allowing them to check part compatibility and place orders even before visiting the shop.

When - The compatibility checker and product information features are available to customers at any time, allowing them to check part compatibility at their convenience. The administrative features of the system, which include inventory management, point of sales transaction recording, repair tracking, pre-order management, and report generation, are used by the shop owners during regular operating hours from 6:00 AM to 6:00 PM daily. Stock monitoring alerts run continuously so the owners are notified right away whenever a product falls below its set stock level.

How: Currently, Edmar Motorcycle Parts and Accessories manages all operations manually, using handwritten logbooks for inventory, hand-recorded sales, memory or printed catalogs for compatibility questions, and informal handling of pre-orders, all of which lead to recurring problems such as stock shortages, recording errors, and untracked orders. MOTOTRACK will streamline these processes through:

Inventory Management – The owners can add, update, and organize all product information, quantities, and details in a centralized digital record, replacing the handwritten logbooks and physical counts currently used by the shop.

Real-Time Stock Monitoring – The owners can monitor stock levels in real time and receive automatic low-stock alerts whenever a product falls below its set level, eliminating the risk of undetected stock shortages.

Parts Compatibility Checker – Customers can check whether a specific motorcycle part is compatible with their make and model directly through the system, removing the need to rely on memory or printed catalogs.

Point of Sales Transaction Recording – The system automatically records every completed sales transaction and updates inventory levels accordingly, replacing handwritten entries with accurate and organized digital records.

Repair and Service Request Tracking – The system tracks ongoing and pending repair and service requests, including the parts used and expected completion time, replacing informal and unorganized manual tracking.

Pre-Order Management – The owners can process and track pre-orders along with down payment details in an organized manner, ensuring no orders are missed or left unrecorded.

Report Generation – The system generates business reports that give the owners a clear overview of sales, inventory, and operations, supporting better and faster decision-making.

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APPENDICES

APPENDIX A.RESOURCE PERSONS

RESOURCE PERSONS

NAME	POSITION	COMPANY/ ORGANIZATION
Ceejay Fajardo	Capstone Project Adviser	STI College Sta. Maria
Denver Castillo	Capstone Project Coordinator	STI College Sta. Maria
Edmar Contreras	Owner	Edmar Motorcycle Parts and Accessories

Table 5.0 Resource Person

APPENDIX B. DIAGRAMS

FLOWCHART

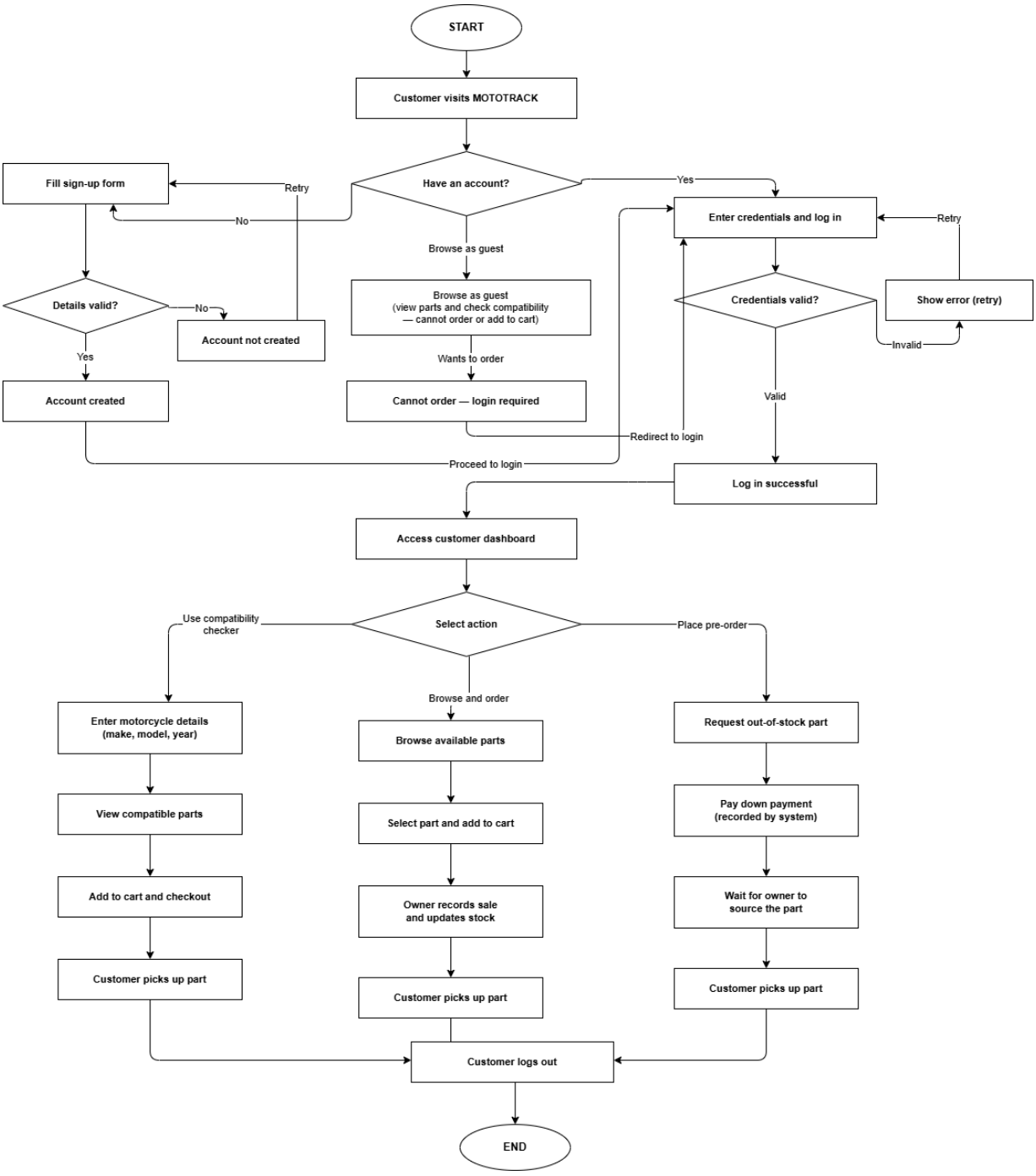


Figure 3.0 Flowchart: Customer Level

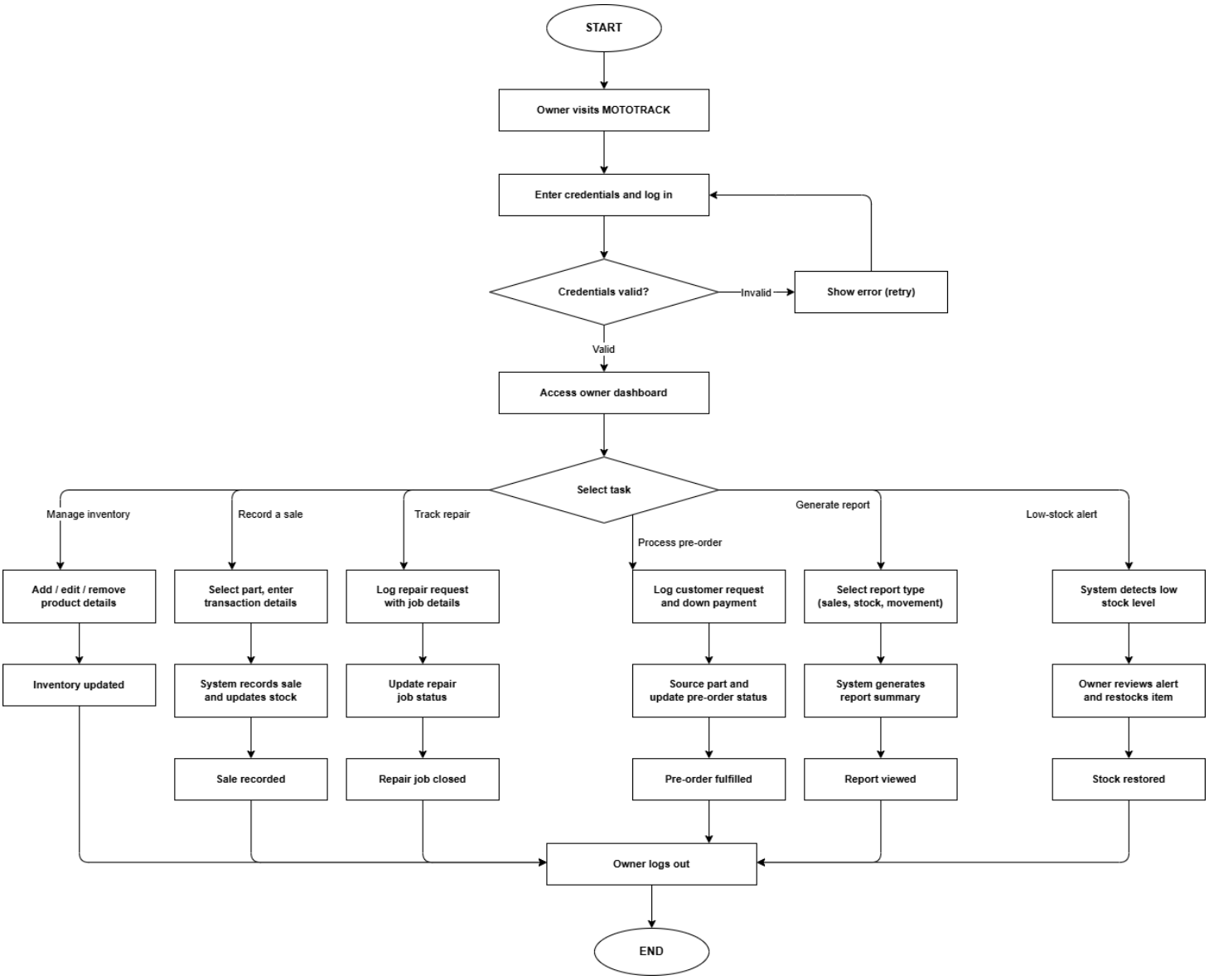


Figure 4.0 Flowchart: Owner Level

CONTEXT DIAGRAM

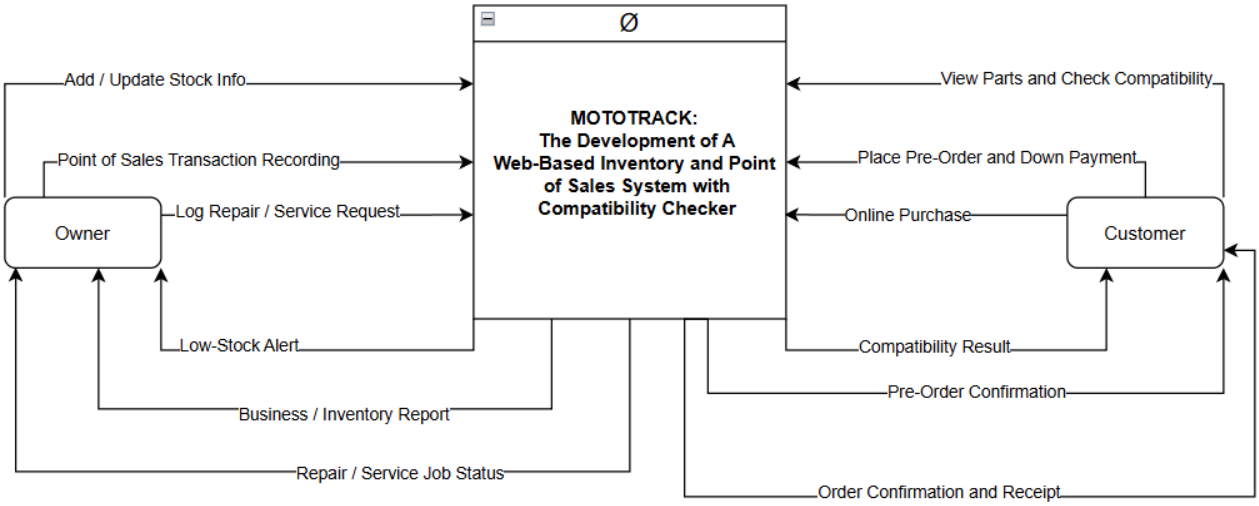


Figure 5.0 Context Diagram

DATA FLOW DIAGRAM

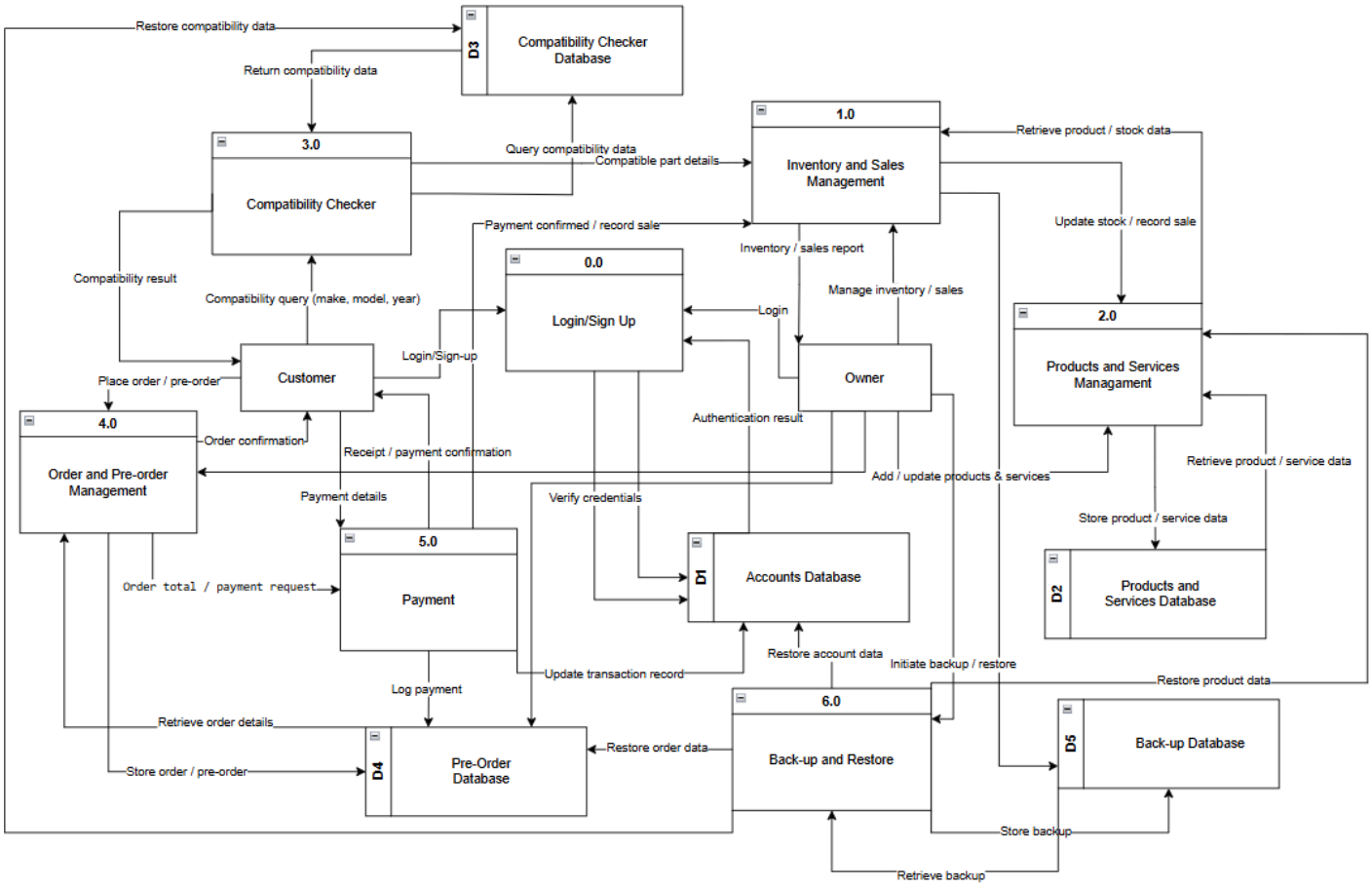


Figure 6.0 Data Flow Diagram

ENTITY RELATIONSHIP DIAGRAM

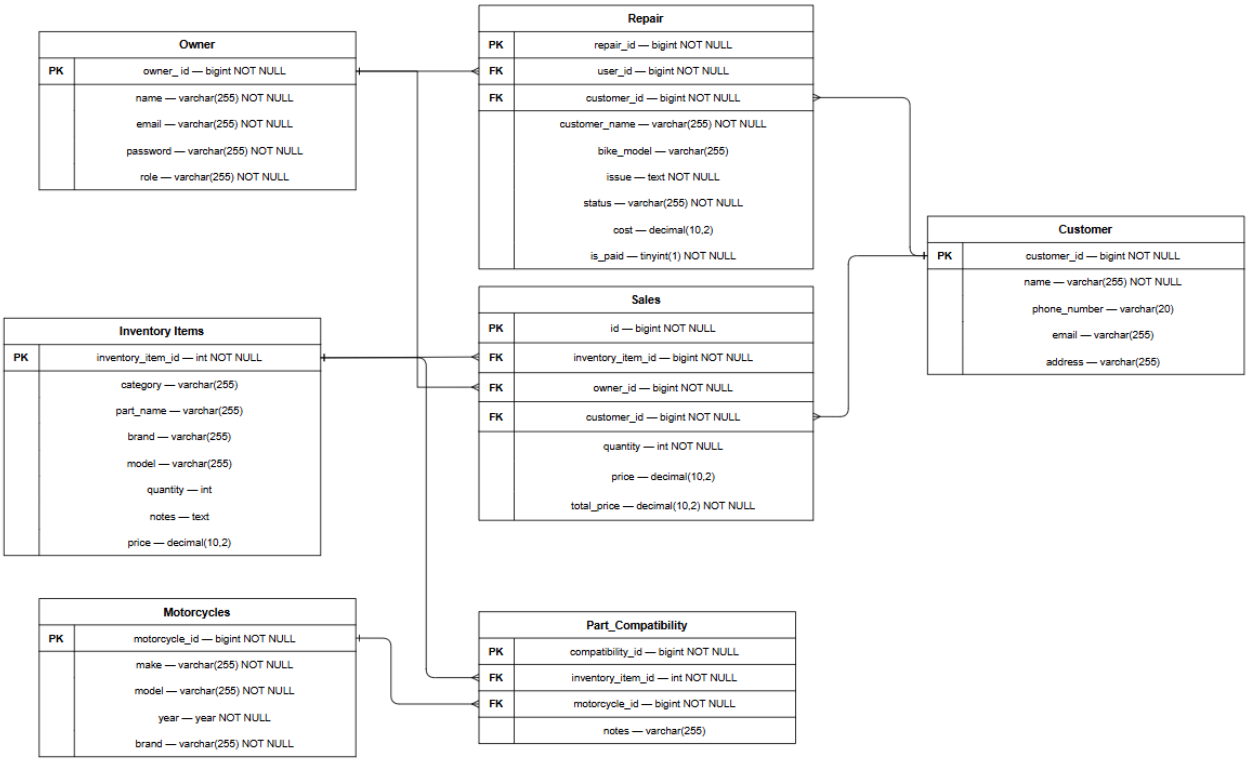


Figure 7.0 Entity Relationship Diagram

APPENDIX C. PERSONAL TECHNICAL VITAE

Curriculum Vitae of

Armina C. Agres

Atlantica Mapulang Lupa, Pandi Bulacan
arminaagres01@gmail.com
09628406858



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	September 2023 - Present	STI College Sta. Maria
Vocational/Technical	June 2021- May 2023	San Jose Academy of Bulacan, INC
High School	June 2018 - March 2021	Jose Abad Santos High School
Elementary	June 2013 - March 2018	Pedro Guevara Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
August 2022	Online Seller	Valenzuela-based TikTok Shop
March 2021	E-commerce Packer	Shopee Seller, Sunprime Tower, Manila
April 2018	Sales Lady & Oncall Cleaner	Divisoria Mall/ Binondo, Manila

SKILLS

SKILLS	Level of Competency	Date Acquired
System Integration & Architecture	Basic	September 2025
Java Fundamentals	Basic	June 2024

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
April 2026	When AI Learns to Talk: Understanding NLP & Cyber ethics and Responsible Use of Technology
February 2026	Research Methodology in ICT: From Concept to Implementation

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Curriculum Vitae of

Clarenz Louiez P. Contreras

Hulo Street Bagong Barrio, Pandi, Bulacan
cclarenzlouiez@gmail.com
09567333571



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	September 2023 - Present	STI College Sta. Maria
Vocational/Technical	August 2021 - June 2023	College of Mary Immaculate. of Pandi, Bulacan, Inc.
High School	June 2020 - July 2021	Alexis G. Santos National High School
	June 2019 - March 2020	Immaculate Conception Child Development Center
	June 2017 - April 2019	San Isidro Parochial School
Elementary	June 2013 - April 2017	San Isidro Parochial School
	June 2011 - April 2013	Christian Academy of Pandi

SKILLS

SKILLS	Level of Competency	Date Acquired
System Integration & Architecture	Basic	September 2025
Java Fundamentals	Basic	June 2024

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
April 2026	When AI Learns to Talk: Understanding NLP & Cyber ethics and Responsible Use of Technology
February 2026	Research Methodology in ICT: From Concept to Implementation



Curriculum Vitae of

Eivann F. Macapagal

Pulong Buhangin, Santa Maria, Bulacan
macapagal.eivan@gmail.com
09271237974

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	September 2023 - Present	STI College Sta. Maria
Vocational/Technical	June 2019 - May 2021	Immaculate Conception PolyTechnic
High School	June 2015 - March 2019	Colegio de Sta. Monica de Angat
Elementary	June 2010 - March 2015	Colegio de Sta. Monica de Angat

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
December 2024	Delivery Rider	Food Panda Taguig City, Metro Manila

SKILLS

SKILLS	Level of Competency	Date Acquired
System Integration & Architecture	Basic	September 2025
Java Fundamentals	Basic	June 2024

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
February 2026	Research Methodology in ICT: From Concept to Implementation